The Central Limit Theorem

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Introduction

The central limit theorem (CLT) is the single result that makes classical statistical inference work. It says that when you draw samples of size n from a population with mean μ and finite variance σ^2, the distribution of the *sample mean* approaches a normal distribution with mean μ and variance σ^2/n as n grows – and this is true regardless of the shape of the original population. Almost every t-test, z-test, and confidence interval for a mean rests on this theorem. Without it, the normal distribution would have no privileged role; with it, the normal is the attractor toward which sample means are pulled.

Prerequisites

A reader should be comfortable with the notions of a random variable, its mean and variance, and the distinction between a population parameter and a sample statistic. Familiarity with R's vectorised operations is helpful but not required.

Theory

Let $X_1, X_2, \ldots, X_n$ be independent, identically distributed random variables with finite mean μ and finite variance σ^2. Define the sample mean $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. The CLT states that as $n \to \infty$,

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2),$$

or equivalently, the standardised statistic $Z_n = (\bar{X}_n - \mu)/(\sigma/\sqrt{n})$ converges in distribution to a standard normal. In practical terms, the sampling distribution of $\bar{X}_n$ is approximately $\mathcal{N}(\mu, \sigma^2/n)$ once n is "large enough". What counts as large enough depends on the skewness of the source: a uniform distribution is nearly normal after $n \approx 5$, a right-skewed distribution like the exponential needs $n \approx 30$, and a very heavy-tailed distribution like the Cauchy (which has no finite mean) is never approximated by the CLT because the assumptions fail.

Assumptions

The theorem requires three conditions:

1. Observations are independent of each other.
2. Observations are identically distributed – they come from the same population.
3. The population has a finite variance.

Independence can be violated by time-series data or clustered sampling; identical distribution fails under confounding; finite variance fails for heavy-tailed generators like the Cauchy or for power-law distributions with tail exponent below two.

R Implementation

The clearest way to see the CLT is by simulation. We draw many samples from a skewed population, compute the mean of each, and plot the histogram of those means.

```r
library(ggplot2)
set.seed(2026)

simulate_clt <- function(n, reps = 10000, rate = 1) {
  replicate(reps, mean(rexp(n, rate = rate)))
}

means_n5   <- simulate_clt(n = 5)
means_n30  <- simulate_clt(n = 30)
means_n100 <- simulate_clt(n = 100)

df <- data.frame(
  mean_value = c(means_n5, means_n30, means_n100),
  n = factor(rep(c(5, 30, 100), each = 10000),
             levels = c(5, 30, 100))
)

ggplot(df, aes(x = mean_value)) +
  geom_histogram(aes(y = after_stat(density)),
                 bins = 60, fill = "steelblue", colour = "white") +
  facet_wrap(~ n, scales = "free",
             labeller = labeller(n = function(x) paste0("n = ", x))) +
  stat_function(fun = dnorm,
                args = list(mean = 1, sd = 1 / sqrt(as.numeric(levels(df$n)[1]))),
                colour = "red", linewidth = 1) +
  labs(x = expression(bar(X)), y = "Density") +
  theme_minimal(base_size = 12)
```

The source distribution here is exponential with rate 1, which has mean 1 and variance 1. At $n = 5$ the histogram is still visibly right-skewed; by $n = 30$ it is symmetric and bell-shaped; by $n = 100$ it is indistinguishable from the theoretical normal curve.

Output & Results

Running the simulation produces three panels. The empirical standard deviation of the sample means in each panel is close to $1/\sqrt{n}$: about 0.447 at $n = 5$, 0.183 at $n = 30$, and 0.100 at $n = 100$. The empirical mean is close to 1 in every panel, as the CLT requires.

Interpretation

For a manuscript, the CLT is rarely cited by name; instead it justifies the use of a normal-based confidence interval or a t-test when the raw data are not normal but the sample size is moderate. A reader reports "with $n = 50$ the sampling distribution of the mean is well-approximated by a normal, so we use a t-based 95% confidence interval", not "by the central limit theorem". The theorem is background; the application is the CI.

Practical Tips

- For strongly skewed populations (income, biomarker concentrations, counts), plan for $n \geq 30$ before relying on CLT-based inference.
- The CLT applies to the *mean*, not to individual observations: a single new observation is not made approximately normal by a large n.
- When variance is suspected to be infinite or very heavy-tailed, use a robust location estimator (median, trimmed mean) or a bootstrap confidence interval.

- The convergence rate is governed by the Berry-Esseen theorem, which bounds the distance to the normal in terms of n and the third absolute moment.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Scientific process and research workflow
- R, RStudio, Quarto, and renv toolchain
- Data types, tidy data, accuracy and precision
- Import, joins, and missingness with dplyr